

面向网络评论的观点信息抽取研究

Mining Opinion Targets and Opinion Words from Online Reviews

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Opinion Mining

- An *opinion* is a quintuple (Liu 2006)

$$(e_j, f_{jk}, so_{ijkl}, h_i, t_l)$$

- e_j is a target entity.
 - f_{jk} is a feature of the entity e_j .
 - so_{ijkl} is the sentiment value of the opinion
 - h_i is an opinion holder.
 - t_l is the time when the opinion is expressed.
- Opinion Mining: Subjective Information Extraction
 - Unstructured → Structured

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Apple iPhone 8 GB (first generation) (AT&T - GSM) from Apple in Mobile Phones

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 **\$500 new, \$180 used** from [4 sellers](#)

★★★★☆ 1,784 reviews

Reviews

Summary - Based on 1,784 reviews

1 2 3 4 stars 5 stars

"Easy texting and buttons are good size to heat." "Awful speaker though."
"The photos have good clarity and is very handy." "With its touch screen it makes life so much easier."
"You can download music, videos, and movies (love)" "The worst thing about the iphone is the price."

Apple iPhone - 8GB (AT&T) Review

★★★★☆ By Kent German - Jun 30, 2007 - Editorial review - CNET

Pros: The Apple iPhone has a stunning display, a sleek design, and an innovative multitouch user interface. Its Safari browser makes for a superb Web surfing experience, and it offers easy-to-use apps. As an iPod, it shines.

Cons: The Apple iPhone has variable call quality and lacks some basic features found in many cell phones, including stereo Bluetooth support and 3G compatibility. Integrated memory is stingy for an iPod, and you have to sync the iPhone to manage music content.

Bottom Line: Despite some important missing features, a slow data network, and call quality that doesn't always deliver, the Apple iPhone sets a new benchmark for an integrated cell phone and MP3 player.

Editor's note (7/11/2008): This is the review of the original first-generation iPhone model, released June 2007. Coverage of the 3G iPhone model released July 11, 2008 is available [here](#).

Photo gallery: Apple iPhone

Editor's note: Apple eliminated the 4GB model on September 5, 2007, two months after the iPhone's initial

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Apple iPhone 3GS 16GB - smartphone - WCDMA (UMTS) / GSM

\$449 and up (6 stores)

★★★★★ User reviews (285)

★★★★★ Expert reviews (1)

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USER REVIEWS



SCORECARD: EASE OF USE ([See all](#))

87 positive reviews | no negative comments

Email Emailing is very basic, easy to use and works brilliantly.

★★★★★ KaylieLandymore · 10/6/2010 · www.ciao.co.uk [see all](#)

Pros: good maps, memory, quick, easy to use, access to loads of different features

★★★★★ anklechris · 9/30/2010 · www.ciao.co.uk [see all](#)

Pros: Has everything, brilliant music player and camera, great for games, easy to use.

★★★★★ doodles12 · 5/24/2010 · www.ciao.co.uk [see all](#)

Safari: This is the Iphone Web browser, its very easy to use, very quick and you can go wrong with it, it has a "Bookmarks" Section so can store all your favorites websites in specific folders, to can navigate forwards to backwards on the web...

★★★★★ alanh087 · 8/28/2010 · www.ciao.co.uk [see all](#)

The phone is very easy to use, everything is very intuitive.

★★★★★ Melski1979 · 9/20/2010 · www.ciao.co.uk [see all](#)

SCORECARD

Ease of use(87)

Camera(86)

Screen(55)

Wireless Interface(44)

Music Player(44)

Functionality(44)

Power Supply(40)

Durability(31)

Reception(25)

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店铺动态评分 与同行业相比
描述相符: 4.6 低于 2.58%
服务态度: 4.6 低于 1.21%
发货速度: 4.7 低于 0.52%

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关键字

价格 -

[Q](#)

客服中心

工作时间
周一至周五: 9:00-0:30
周六至周日: 9:00-0:30

商品详情

包装和参数

累计评价 2053

月成交记录 135件

电器城服务详情

与描述相符 4.6

性价比很高(199) 屏幕不错(136) 系统流畅(123) 外观靓丽(103)

电池耐用(88) 性能好(53) 电池一般(280) 通话质量一般(70) 屏幕一般(37)

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拿到手,发现屏幕里有好多黑色斑点,以为是屏幕有问题,把贴膜撕下来后发现不是屏幕的问题,可是贴膜就废了!既然送贴膜就贴好点啊!搞的我以为是屏幕有问题!其他还好,性价比高,就是电池不够用,总体还行,天语总部发的货,13年9月23号的新机,客服服务很到位,给个好评吧!

解释:祝您购物愉快!!为了保护屏幕,手机出厂时自带的都是简装出厂膜哦 感谢您的好评!!亲的优质评价是对我们最好的宣传,我们一定会更加努力做到更好(*^__^*)使用中有任何问题随时都可以联系本店在线客服(工作时间:9:00—00:30),客服会及时为您解决。

10.31

机身颜色: 黑色
手机套餐: 官方标配

yuyelei91
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tb226342752

Opinion Summarization

- “我今年入手诺基亚5800，把玩不到24小时，目前感觉5800屏幕很好，操作也很方便，通话质量也不错，但是外形有些偏女性化，不适合男生。这些都是小问题，最主要的问题是电池不耐用，只能坚持一天，反正我觉得对不起这个价格。”



三个基本步骤

■ 观点信息的挖掘

- 建立词典（挖掘先验知识）

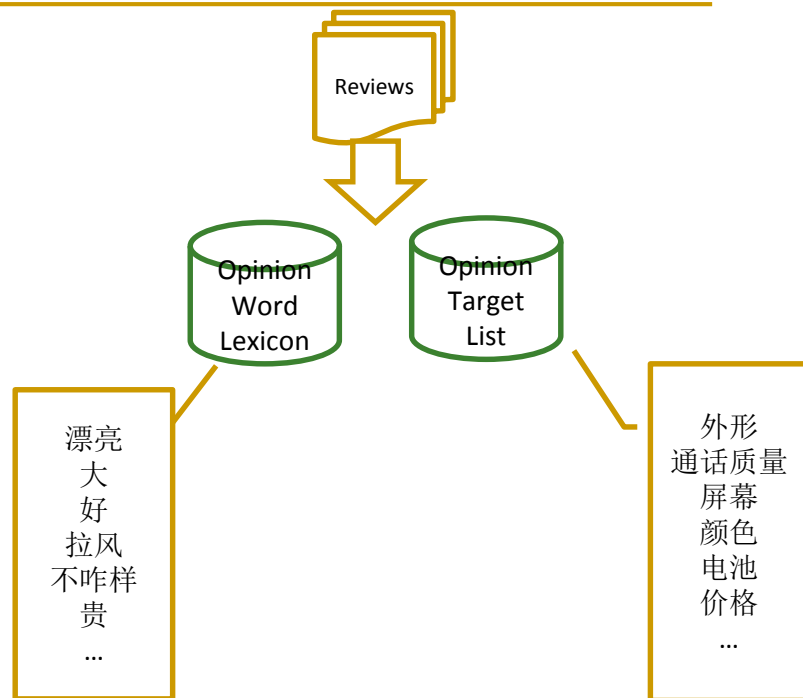
■ 观点要素的识别

- 识别句子中观点评价对象以及观点评价词的mention

- 服务真赞

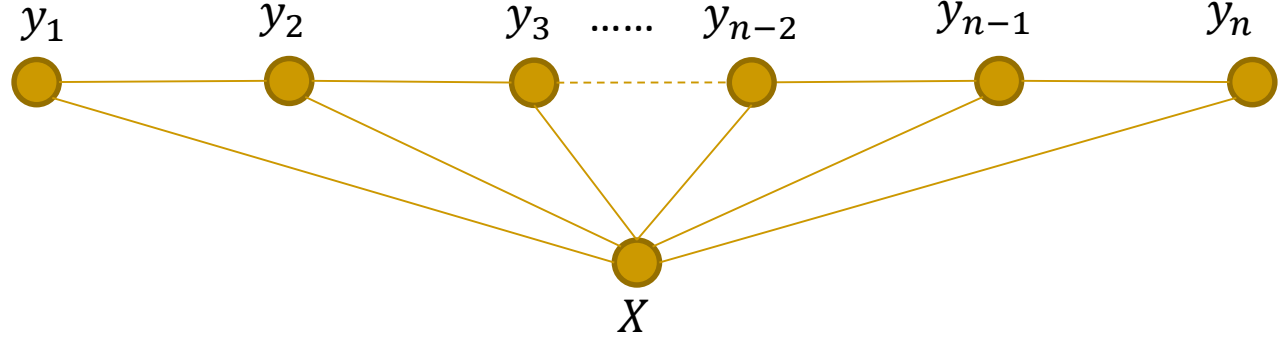
■ 观点倾向的判别

- 但是价格偏贵，人均消费100块。(Negative)
- 服务真赞。(Positive)



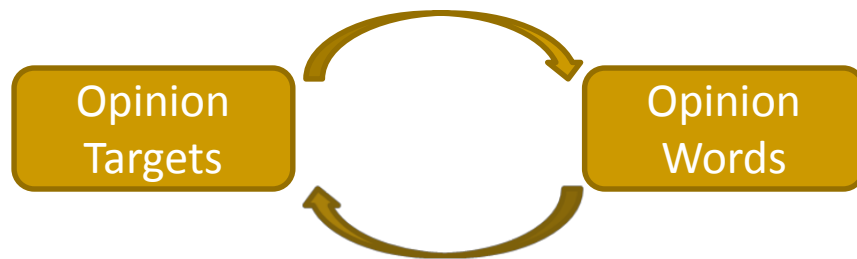
Outline

- Previous work
 - Mining Opinion Targets/Words with Syntactic Information
 - Mining Opinion Target with Word Alignment Model
 - Conclusions
-

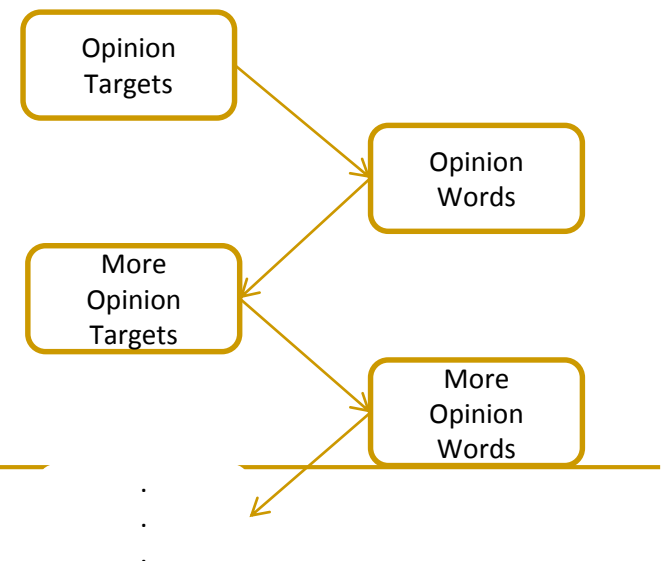


Pervious Work (Cont.)

- Unsupervised Methods (Hu 2004, Popescu 2005, Qiu 2010, Zhang 2010)
 - Aim : generating the lists of opinion targets and opinion words
 - Opinion target: Nouns or Noun phrases
 - Opinion words: Adjectives or Verbs
 - Opinion words usually modify opinion targets
 - Bootstrapping framework



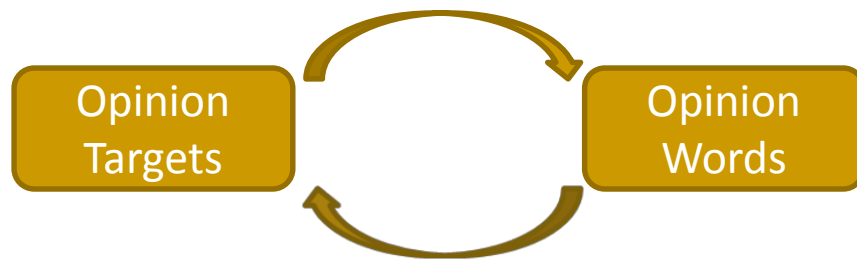
Double Propagation



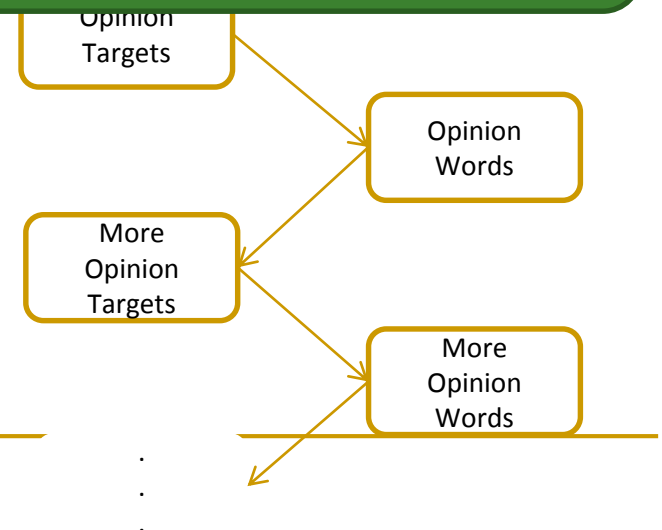
Pervious Work (Cont.)

- Unsupervised Methods (Hu 2004, Popescu 2005, Qiu 2010, Zhang 2010)
 - Aim : generating the lists of opinion targets and opinion words
 - Opinion target: Nouns or Noun phrases

Key: identifying the modified relations between opinion target and opinion word (Opinion Relation)



Double Propagation



Pervious Work (Cont.)

- Adjacent Method (Hu 2004, Wang 2008)

这款 手机的外观 很 漂亮

- It's difficult to capture these opinion relations

这 款 漂亮 的 新 手机 具有 让人 惊叹 的 大 屏幕



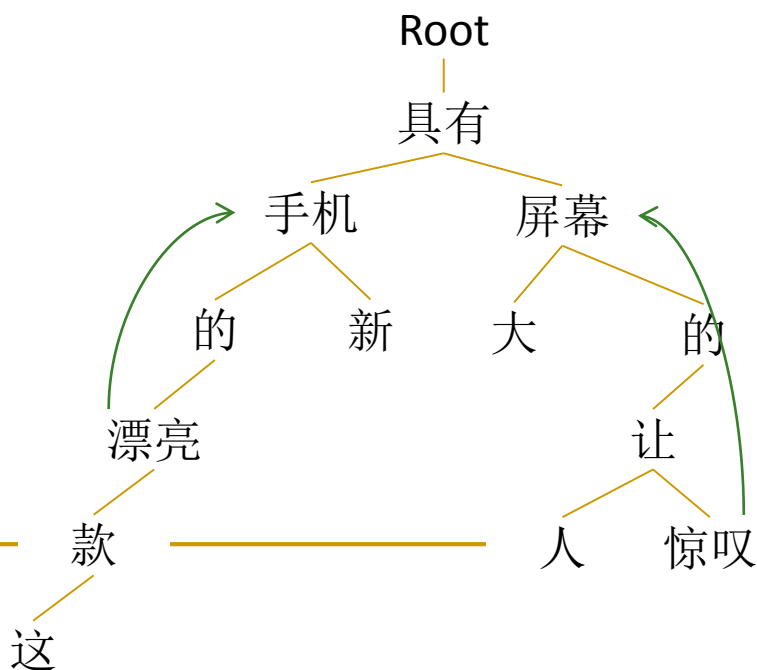
这 家 小店 位于 小巷 的 深处 ， 尽管 位置 不 好 找 ， 但是 仍然 很 赞



Pervious Work (Cont.)

- Syntax-based Method: Popescu (EMNLP 2005), Qiu (IJCAI 2009)
 - Employing the dependency relations between opinion targets and opinion words

这 款 **漂亮** 的 新 **手机** 具有 让人 **惊叹** 的 **大** 屏幕



Typed dependencies

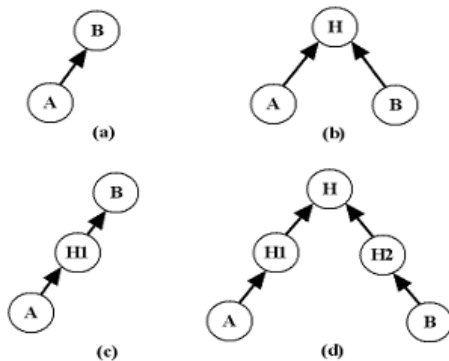
```
det(款-2, 这-1)
nsubj(漂亮-3, 款-2)
dep(的-4, 漂亮-3)
rcmod(手机-6, 的-4)
amod(手机-6, 新-5)
nsubj(具有-7, 手机-6)
root(ROOT-0, 具有-7)
dep(的-11, 让-8)
dobj(让-8, 人-9)
dep(让-8, 惊叹-10)
rcmod(屏幕-13, 的-11)
amod(屏幕-13, 大-12)
dobj(具有-7, 屏幕-13)
```

Pervious Work (Cont.)

■ Popescu (EMNLP 2005)

Extraction Rules	Examples
$\text{if } \exists(M, NP = f) \rightarrow po = M$	(expensive) scanner
$\text{if } \exists(S = f, P, O) \rightarrow po = O$	lamp has (problems)
$\text{if } \exists(S, P, O = f) \rightarrow po = P$	I (hate) this scanner
$\text{if } \exists(S = f, P, O) \rightarrow po = P$	program (crashed)

■ Qiu (IJCAI 2009, CL 2010)



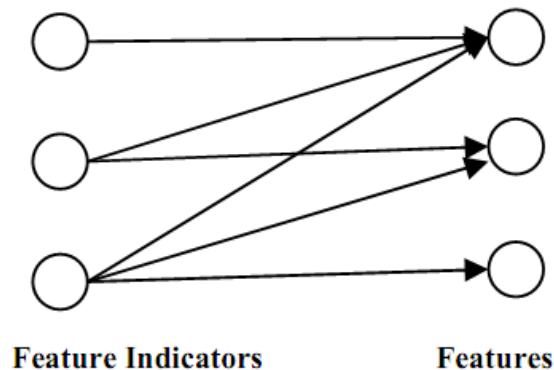
	Observations	Constraints	Outputs
R1 ₁	$S_{i0} \rightarrow S_{i0} \text{-Dep} \rightarrow S_{j0}$	$S_{j0} \in \{S\},$ $S_{i0} \text{-Dep} \in \{CONJ\},$ $POS(S_{i0}) \in \{JJ\}$	$s = S_{i0}$
R1 ₂	$S_i \rightarrow S_i \text{-Dep} \rightarrow H \leftarrow S_i \text{-Dep} \leftarrow S_j$	$S_j \in \{S\},$ $S_i \text{-Dep} = S_j \text{-Dep},$ $POS(S_j) \in \{JJ\}$	$s = S_j$
R2 ₁	$S \rightarrow S \text{-Dep} \rightarrow F$	$F \in \{F\},$ $S \text{-Dep} \in \{MR\},$ $POS(S) \in \{JJ\}$	$s = S$
R2 ₂	$S \rightarrow S \text{-Dep} \rightarrow H \leftarrow F \text{-Dep} \leftarrow F$	$F \in \{F\},$ $S/F \text{-Dep} \in \{MR\},$ $POS(S) \in \{JJ\}$	$s = S$
R3 ₁	$S \rightarrow S \text{-Dep} \rightarrow F$	$S \in \{S\},$ $S \text{-Dep} \in \{MR\},$ $POS(F) \in \{NN\}$	$f = F$
R3 ₂	$S \rightarrow S \text{-Dep} \rightarrow H \leftarrow F \text{-Dep} \leftarrow F$	$S \in \{S\},$ $S/F \text{-Dep} \in \{MR\},$ $POS(F) \in \{NN\}$	$f = F$
R4 ₁	$F_{i0} \rightarrow F_{i0} \text{-Dep} \rightarrow F_{j0}$	$F_{j0} \in \{F\},$ $F_{i0} \text{-Dep} \in \{CONJ\},$ $POS(F_{i0}) \in \{NN\}$	$f = F_{i0}$
R4 ₂	$F_i \rightarrow F_i \text{-Dep} \rightarrow H \leftarrow F_i \text{-Dep} \leftarrow F_j$	$F_j \in \{F\},$ $F_i \text{-Dep} = F_j \text{-Dep},$ $POS(F_j) \in \{NN\}$	$f = F_j$

Pervious Work (Cont.)

■ Zhang (COLING 2010)

- Designing more syntactic patterns
 - NP + Prep + CP: “battery of the camera”
 - NP CP or CP NP: “mattress pad”
 - “no” Pattern: “no noise”

□ HITS Algorithm (improving precision)



$$S = S(f) \log (\text{freq}(f))$$

Problems

- Adjacent Method:
 - Difficulty to Identify the long-span opinion relations
- Syntax-based Method:
 - The performance of parsing! Reviews are usually informal texts!
 - How to avoid parsing errors?
 - Can we don't use parsing?
- Error Propagation in Bootstrapping Framework
 - How to avoid error propagation?

Outline

- Previous work
 - Mining Opinion Targets/Words with Syntactic Information
 - Mining Opinion Target with Word Alignment Model
 - Conclusions
-

Mining Opinion Targets/Words with Syntactic Information

- Previous syntactic patterns is often manually designed
- Which patterns is more suitable?
- Incorporate the syntactic patterns confidence into extraction

Step1: Walk on graph

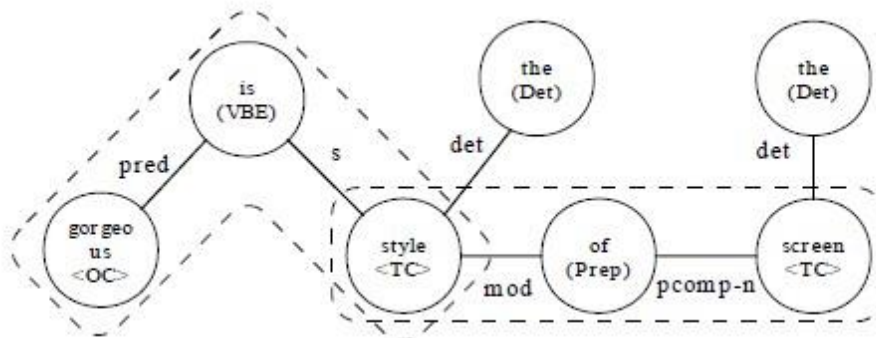
- False opinion targets and opinion words
 - nice people, good thing
- Long-tail targets

Step2: Learn to refine

Step1: Walk

- Automatically learn syntactic patterns
 - Seeds (Opinion words):
 - good, bad, colorful, gorgeous.....
 - Shortest path in dependency tree
 - Filtering infrequent patterns

The style of the screen is gorgeous



<OC>{pred}(VBE){s}<TC>

<TC>{mod}(Prep){pcompn}<TC>

Step1: Walk (Cont.)

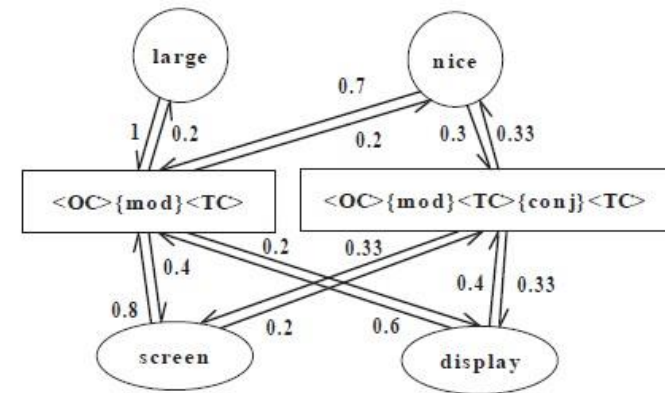
■ Constructing a Sentiment Graph

□ Three Vertices

- Opinion Targets Candidates
- Opinion Words Candidates
- Used Syntactic Patterns

□ Assumption

- If a word (noun/noun phrase) is an opinion targets, the word (adjective/verb) which modifies this noun has higher probability to be an opinion word, and vice versa.
- If an opinion word/target is with higher confidence, the syntactic patterns containing this term are more likely to be a good pattern, and vice versa.



Step1: Walk (Cont.)

- Estimating Each Candidate Confidence using Random Walking Algorithm

$$\mathbf{c}_p^{t+1} = \mathbf{M}_{oc-p}^T \times \mathbf{c}_{oc}^t + \mathbf{M}_{tc-p}^T \times \mathbf{c}_{tc}^t$$

$$\mathbf{c}_{oc}^{t+1} = (1 - \lambda) \mathbf{M}_{p-oc}^T \times \mathbf{c}_p^t + \lambda \mathbf{c}_{oc}^0$$

$$\mathbf{c}_{tc}^{t+1} = \mathbf{M}_{p-tc}^T \times \mathbf{c}_p^t$$

$$\mathbf{M}_{oc-p}(v_o, v_p) = w(v_o, v_p)$$

$$\mathbf{M}_{tc-p}, \mathbf{M}_{p-oc}, \mathbf{M}_{p-tc}$$

$$w(v_a, v_b) = freq(v_a, v_b) / freq(v_a)$$

Step2: Learn

- Refining Results using Self-Learning
 - In target list
 - “nice people”, “good thing”
 - In opinion word list
 - “every time”, “many people”
- Refining targets using Self-Learning Strategy
- Refining opinion words using target results

Refining Targets using Self-Learning Strategy

- How to generate the initial labeled data
 - Noises : domain independent general noun (GN corpora)
 - 1000 most frequent nouns in Google 1-gram
 - Top three levels of hyponyms in WordNet: object, person, group
 - Chinese: Hownet
 - Training data:
 - +1: N most confident targets extracted in Step1, but not occur in GN
 - -1: Other N terms from GN corpora which are also top ranked in the target list

Refining Targets using Self-Learning Strategy (Cont.)

- How to represent each target candidate

$$\mathbf{x}_i = (o_1, \dots, o_n, p_1, \dots, p_m)^T$$

$$x(o_j) = \text{conf}(o_j) \times \frac{\sum_{p_k} \text{freq}(t_i, o_j, p_k)}{\text{freq}(o_j)}$$

$$x(p_k) = \text{conf}(p_k) \times \frac{\sum_{o_j} \text{freq}(t_i, o_j, p_k)}{\text{freq}(p_k)}$$

- Training the refinement classifier
 - We use Transductive SVM

Opinion Word Refinement

- Use the classified opinion target results to refine the opinion words

$$s(o_j) = \sum_{t_i \in T} \sum_{p_k} \frac{s(t_i) \text{conf}(p_k) \text{freq}(t_i, o_j, p_k)}{\text{freq}(t_i)}$$

Experiments

■ Dataset

- ❑ Customer Review Dataset (CRD)
- ❑ COAE 2008
- ❑ Collect a larger dataset: LARGE

Dataset	Domain	#Sentences	#OW	#OT
Large (English)	Hotel	10,000	434	1,015
	MP3	10,000	559	1,158
COAE08 (Chinese)	Camera	2,075	351	892
	Car	4,783	622	1,179

Data set	Number of reviews	Number of sentences
D1	45	597
D2	34	346
D3	41	546
D4	95	1,716
D5	99	740

■ Evaluation

- ❑ P, R and F1

Baselines

- Hu (Hu AAAI 2004)
 - ❑ Extract opinion targets using adjacent rule
 - ❑ Double Propagation
- DP (Qiu CL 2011)
 - ❑ Manual designed syntactic patterns
 - ❑ Double Propagation
- Zhang (Zhang COLING 2010)
 - ❑ Manual designed syntactic patterns
 - ❑ Graph-based Algorithm (HITS algorithm combined with candidate frequency)

Overall Results

Methods	D1			D2			D3			D4			D5			Avg.
	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F	F
Hu	0.75	0.82	0.78	0.71	0.79	0.75	0.72	0.76	0.74	0.69	0.82	0.75	0.74	0.80	0.77	0.76
DP	0.87	0.81	0.84	0.90	0.81	0.85	0.90	0.86	0.88	0.81	0.84	0.82	0.92	0.86	0.89	0.86
Zhang	0.83	0.84	0.83	0.86	0.85	0.85	0.86	0.88	0.87	0.80	0.85	0.82	0.86	0.86	0.86	0.85
Ours-Stage1	0.79	0.85	0.82	0.82	0.87	0.84	0.83	0.87	0.85	0.78	0.88	0.83	0.82	0.88	0.85	0.84
Ours-Full	0.86	0.82	0.84	0.88	0.83	0.85	0.89	0.86	0.87	0.83	0.86	0.84	0.89	0.85	0.87	0.86

Table 2: Results of opinion target extraction on the *Customer Review Dataset*.

Methods	D1			D2			D3			D4			D5			Avg.
	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F	F
Hu	0.57	0.75	0.65	0.51	0.76	0.61	0.57	0.73	0.64	0.54	0.62	0.58	0.62	0.67	0.64	0.62
DP	0.64	0.73	0.68	0.57	0.79	0.66	0.65	0.70	0.67	0.61	0.65	0.63	0.70	0.68	0.69	0.67
Ours-Stage1	0.61	0.75	0.67	0.55	0.80	0.65	0.63	0.75	0.68	0.60	0.69	0.64	0.68	0.70	0.69	0.67
Ours-Full	0.64	0.74	0.69	0.59	0.79	0.68	0.66	0.71	0.68	0.65	0.67	0.66	0.72	0.67	0.69	0.68

Table 3: Results of opinion word extraction on the *Customer Review Dataset*.

Overall Results

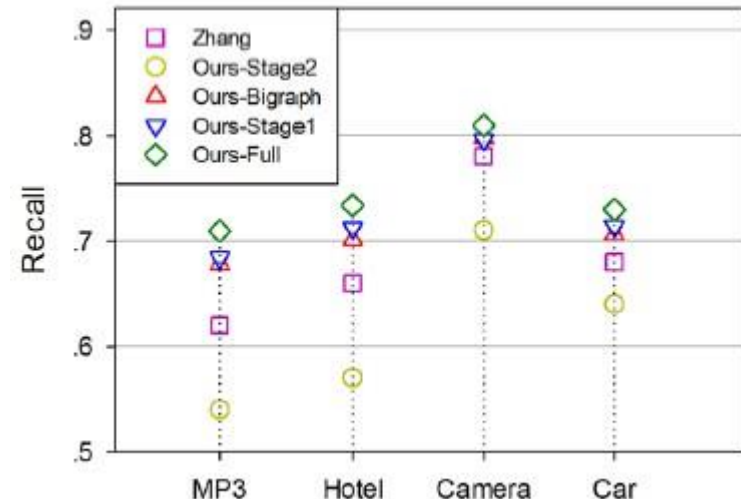
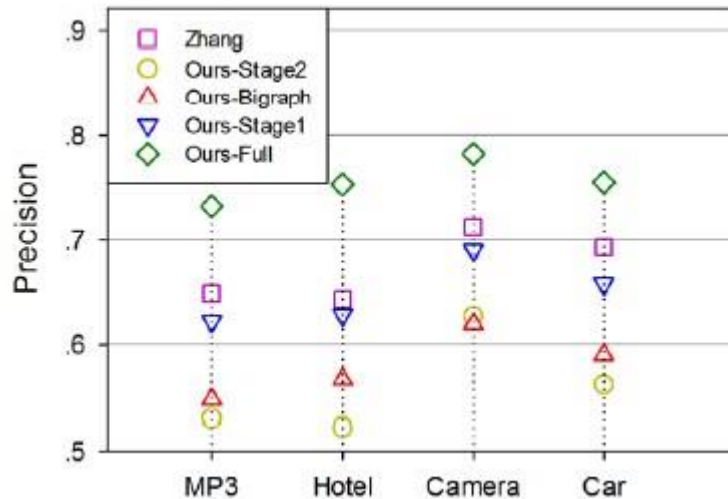
Methods	MP3			Hotel			Camera			Car			Avg.
	P	R	F	P	R	F	P	R	F	P	R	F	F
Hu	0.53	0.55	0.54	0.55	0.57	0.56	0.63	0.65	0.64	0.62	0.58	0.60	0.58
DP	0.66	0.57	0.61	0.66	0.60	0.63	0.71	0.70	0.70	0.72	0.65	0.68	0.66
Zhang	0.65	0.62	0.63	0.64	0.66	0.65	0.71	0.78	0.74	0.69	0.68	0.68	0.68
Ours-Stage1	0.62	0.68	0.65	0.63	0.71	0.67	0.69	0.80	0.74	0.66	0.71	0.68	0.69
Ours-Full	0.73	0.71	0.72	0.75	0.73	0.74	0.78	0.81	0.79	0.76	0.73	0.74	0.75

Table 4: Results of opinion targets extraction on *Large* and *COAE08*.

Methods	MP3			Hotel			Camera			Car			Avg.
	P	R	F	P	R	F	P	R	F	P	R	F	F
Hu	0.48	0.65	0.55	0.51	0.68	0.58	0.72	0.74	0.73	0.70	0.71	0.70	0.64
DP	0.58	0.62	0.60	0.60	0.66	0.63	0.80	0.73	0.76	0.79	0.71	0.75	0.68
Ours-Stage1	0.59	0.69	0.64	0.61	0.71	0.66	0.79	0.78	0.78	0.77	0.77	0.77	0.71
Ours-Full	0.64	0.67	0.65	0.67	0.69	0.68	0.82	0.78	0.80	0.80	0.76	0.78	0.73

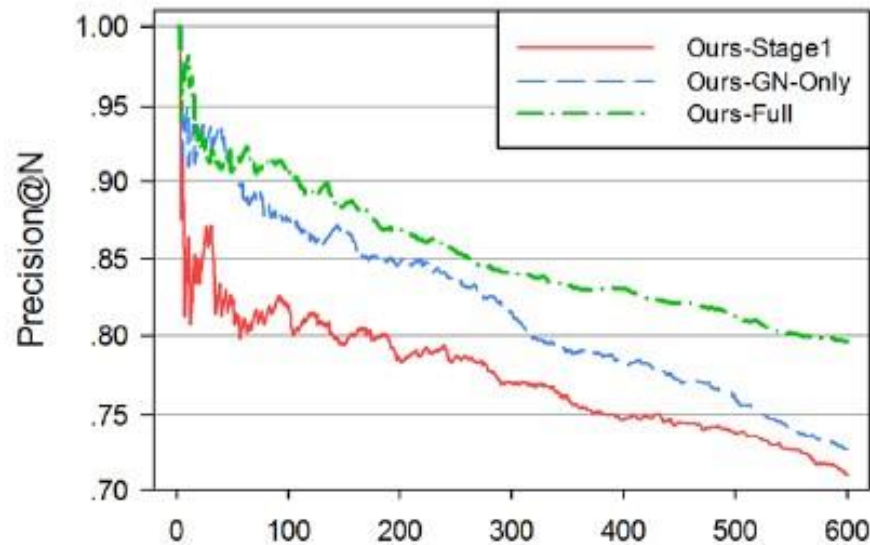
Table 5: Results of opinion words extraction on *Large* and *COAE08*.

The Effectiveness of Our Graph Walking



- Ours-Bigraph: Do not consider the pattern confidence in the Stage1
- Ours-Stage1: Only use stage 1
- Ours-Stage2: Only use stage 2
- Ours-Full: The full implementation of our method

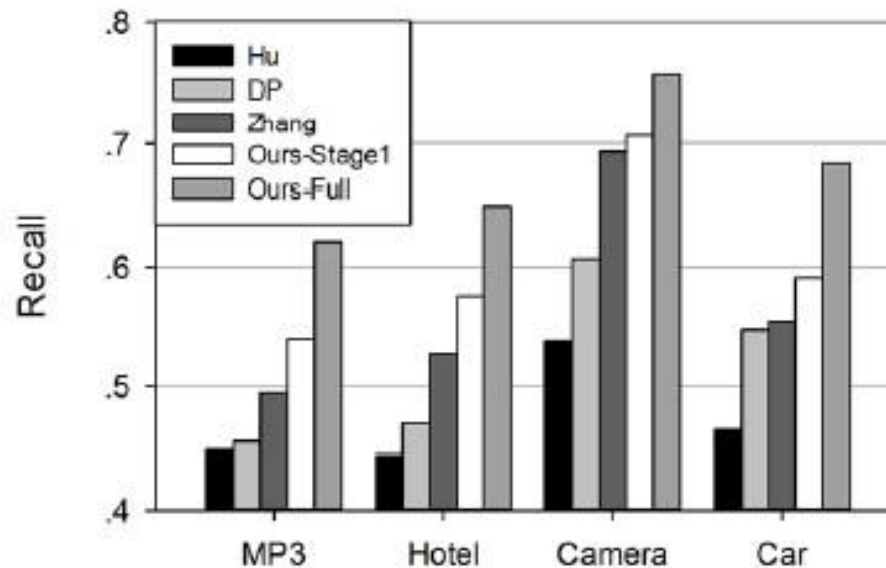
The Effectiveness of Self-Learning



- Ours-Stage1: Do not use stage 2 to refine the results
- Ours-GN-Only: Only use the general noun lexicon to refine the results
- Ours-Full: The full implementation of our method

Detailed Analysis

- Long-tail targets: 60% targets with lower frequency



Outline

- Previous work
 - Mining Opinion Targets/Words with Syntactic Information
 - Mining Opinion Target with Word Alignment Model
 - Conclusions
-

Basic Idea

- Double Propagation (bootstrapping)
 - Error propagation
- Solution: Graph-based ranking algorithm
 - Potential opinion targets: Noun/Noun Phrase
 - Potential opinion words: Adjectives
 - Assumption
 - If an adjective has higher confidence to be an opinion word, the noun/noun phrase it modifies will have higher confidence to be an opinion target.
 - If a noun/noun phrase has higher confidence to be an opinion target, the adjective which modifies it will be highly possible to be an opinion word.

Basic Idea (Cont.)

- Identifying opinion relations in sentences is the key
 - Adjacent methods
 - Syntax-based methods
- Can we do not use parsing for informal texts

Word Alignment

Basic Idea (Cont.)

■ Word alignment

C: 救援 人员 在 倒塌的 房屋 里 寻找 生还者

E: Rescue workers search for survivors in collapsed houses

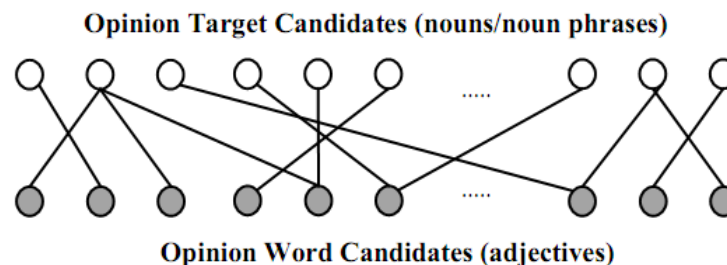
■ Monolingual word alignment

E: The phone has a colorful and even amazing screen

E: The phone has a colorful and even amazing screen

Our solution

- Constructing a bipartite graph
 - Globally estimating each candidate confidence



- Two factors:
 - **Opinion Relevance:** The opinion association between candidate and opinion words
 - **Target Importance:** The salience of a candidate in the corpus

Candidate Confidence Estimation

■ Target Importance:

$$Importance(c) = \frac{tf-idf(c)}{\sum_c tf-idf(c)}$$

Google N-gram
Corpus

■ Opinion Relevance:

- Capturing opinion relations in sentences
- Estimating the associations between words
- We use word alignment model

$$p(w_N | w_A) = \frac{Count(w_N, w_A)}{Count(w_A)}$$

$$\begin{aligned} Association(w_N, w_A) \\ = (t/p(w_N | w_A) + (1-t)/p(w_A | w_N))^{-1} \end{aligned}$$

Capturing opinion relations using WAM

- Replicate sentence to generate sentence pair
- IBM model (IBM-1, IBM-2, IBM-3)

$$P_{IBM-1}(A|S) \propto \prod_{j=1}^n t(w_j | w_{a_j})$$

词之间的
共现关系

$$P_{IBM-2}(A|S) \propto \prod_{j=1}^n t(w_j | w_{a_j}) d(j | a_j, n)$$

词之间的
位置关系

$$P_{IBM-3}(A|S) \propto \prod_{i=1}^n n(\phi_i | w_i) \prod_{j=1}^n t(w_j | w_{a_j}) d(j | a_j, n)$$

一个词的一对
多关系

E: The phone has a colorful and even amazing screen

E: The phone has a colorful and even amazing screen

Associations Computation using WAM

■ Constrains

- ❑ Nouns/Noun Phrases (Adjectives) must align with adjectives (nouns/noun phrases) or null words
- ❑ Other words can only align with themselves

E: The phone has a colorful and even amazing screen
E: The phone has a colorful and even amazing screen

■ Example of association estimation

	battery life	sound	software
wonderful	0.000	0.042	0.000
poor	0.032	0.000	0.026
long	0.025	0.000	0.000

Candidate Confidence Estimation on the graph

- How to model these two factors
 - Solution: Random Walk with Restart

$$C^{t+1} = (1-\lambda) \times \overbrace{M^T \times M}^{\text{Candidate Opinion Relevance}} \times C^t + \lambda \times \overbrace{S}^{\text{Candidate Importance}}$$

$$M = \begin{bmatrix} \vdots & \dots & m \\ \vdots & \ddots & \vdots \\ \vdots & \dots & \vdots \end{bmatrix}$$

Association(w_N, w_A)

$$S = \begin{bmatrix} s \end{bmatrix}$$

Importance(c)

$$C = \lambda \times (I - (1-\lambda) \times M^T \times M)^{-1} \times S$$

Experimental results

Methods	Camera			Car			Laptop			Phone		
	P	R	F	P	R	F	P	R	F	P	R	F
Hu	0.63	0.65	0.64	0.62	0.58	0.60	0.51	0.67	0.58	0.69	0.60	0.64
DP	0.71	0.70	0.70	0.72	0.65	0.68	0.58	0.69	0.63	0.78	0.66	0.72
Zhang	0.71	0.78	0.74	0.69	0.68	0.68	0.57	0.80	0.67	0.80	0.71	0.75
Ours	0.75	0.81	0.78	0.71	0.71	0.71	0.61	0.85	0.71	0.83	0.74	0.78

(a) COAE 2008

Methods	Hotel			MP3			Restaurant		
	P	R	F	P	R	F	P	R	F
Hu	0.60	0.65	0.62	0.61	0.68	0.64	0.64	0.69	0.66
DP	0.67	0.69	0.68	0.69	0.70	0.69	0.74	0.72	0.73
Zhang	0.67	0.76	0.71	0.67	0.77	0.72	0.75	0.79	0.77
Ours	0.71	0.80	0.75	0.70	0.82	0.76	0.80	0.84	0.82

(b) Large

Methods	D1			D2			D3			D4			D5		
	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F
Hu	0.75	0.82	0.78	0.71	0.79	0.75	0.72	0.76	0.74	0.69	0.82	0.75	0.74	0.80	0.77
DP	0.87	0.81	0.84	0.90	0.81	0.85	0.90	0.86	0.88	0.81	0.84	0.82	0.92	0.86	0.89
Zhang	0.83	0.84	0.83	0.86	0.85	0.85	0.86	0.88	0.87	0.80	0.85	0.83	0.86	0.86	0.86
Ours	0.84	0.85	0.84	0.87	0.85	0.86	0.88	0.89	0.88	0.81	0.85	0.83	0.89	0.87	0.88

(c) CRD

Overall performance (Cont.)

■ An Example:

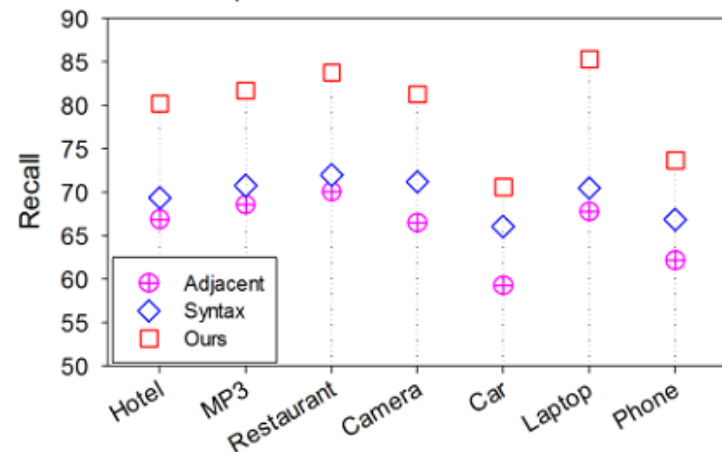
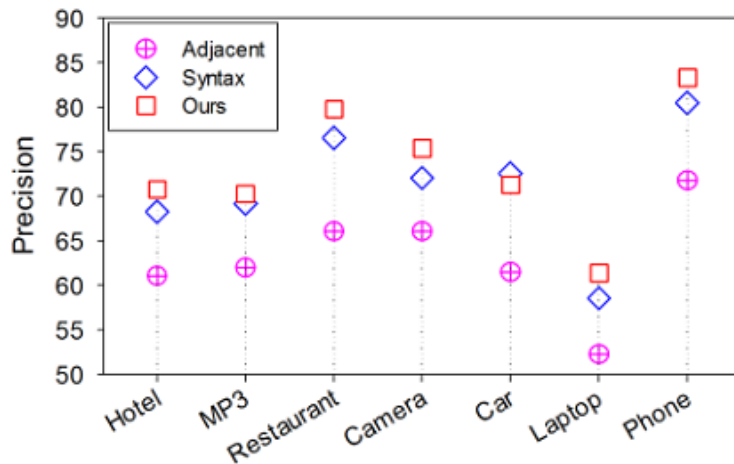
Hu	quality, thing , drive, feature, battery, sound, time , music, price
DP	quality, battery, software, device, screen, file, thing , feature, battery life
Zhang	quality, size, battery life, hour , version, function, upgrade, number , music
Ours	quality, battery life, voice recorder, video, fm radio, battery, file system, screen, lcd screen

WAM is Useful?

■ Compared Methods

- Adjacent: Adjacent rule
- Syntax: Syntactic patterns
- Ours: WAM

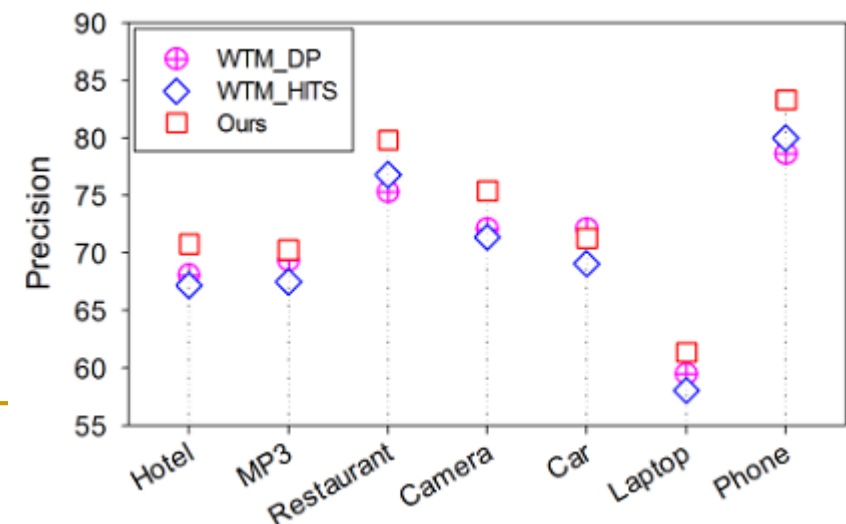
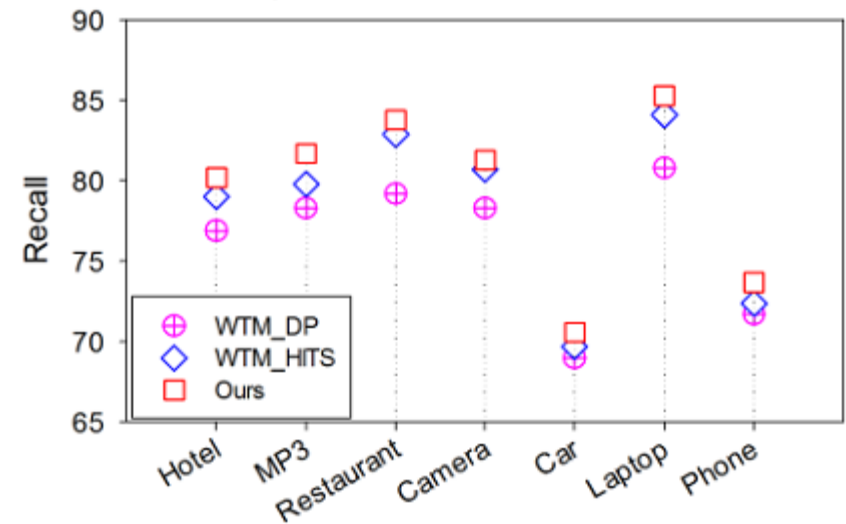
+ Our Graph-based Algorithm



Our Graph-based Method is Useful?

■ Compared Methods

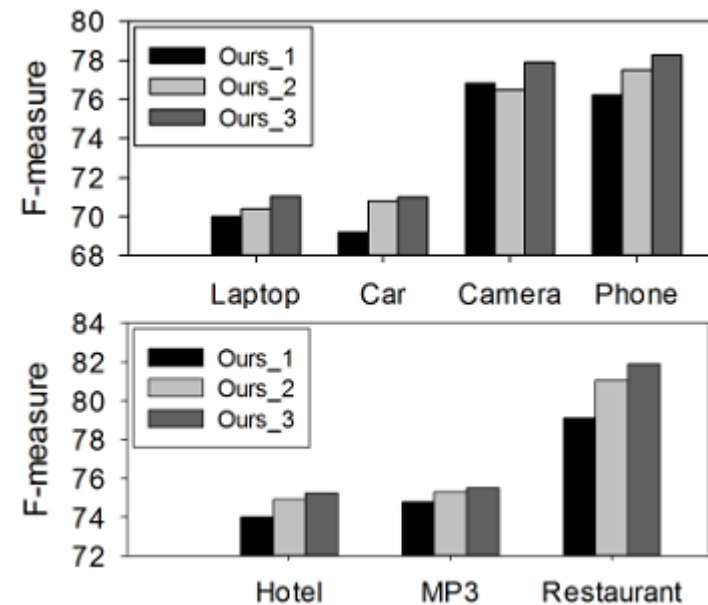
- WTM_DP
 - WAM + Double Propagation
- WTM_HITS
 - WAM + HITS Algorithm
- Ours



Different WAMs

■ Compared Methods:

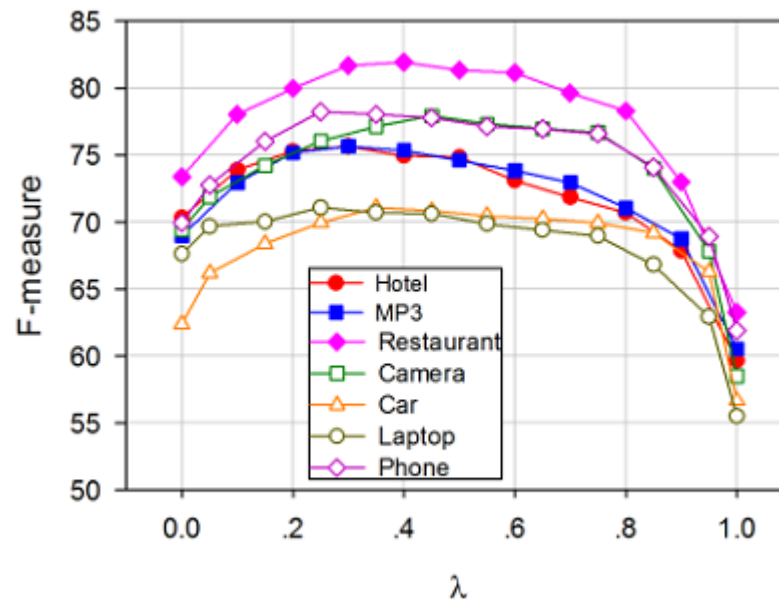
- Ours_1: IBM-1
- Ours_2: IBM-2
- Ours_3: IBM-3



Effect of Parameter λ

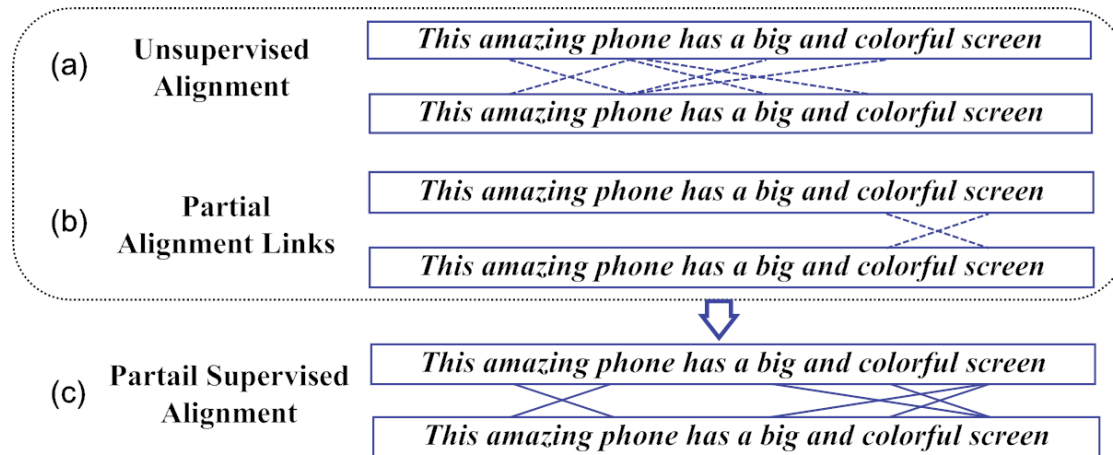
■ Opinion Relevance vs. Candidate Importance

$$C^{t+1} = (1 - \lambda) \times M^T \times M \times C^t + \lambda \times S$$



More Consideration

- Is the used standard word alignment perfect
 - ❑ WAM: training in a completely unsupervised procedure
 - ❑ We need some supervision



More Consideration (Cont.)

- Is standard random walk algorithm enough
 - “beautiful screen”, “beautiful phone”
 - General words: “good”, “bad”.....
 - “good screen”, “good sound”, “good thing”, “good feeling”
 - “good”: High-degree vertex in the bipartite graph
 - “thing”、 “people”
 - Make penalty on these high-degree vertex in the random walking process

Partially Word Alignment Model (Qin Gao, Workshop on SMT, ACL 2010)

- Given the partial alignment links $\hat{A} = \{(i, a_i) | i \in [1, n], a_i \in [1, n]\}$

$$A^* = \underset{A}{\operatorname{argmax}} P(A|S, \hat{A})$$

- Three steps
 - Train a simple alignment model to generate an initial alignments for IBM-3
 - Update the alignments to make it consistent to the partial alignments links
 - Optimize the alignment under the constraints iteratively

$$P(w_i | w_{a_i}) = \begin{cases} \lambda, & A \text{ is inconsistent with } \hat{A} \\ P(w_i | w_{a_i}) + \lambda, & \text{otherwise} \end{cases}$$

Algorithm 1 Constrained Hill-Climbing

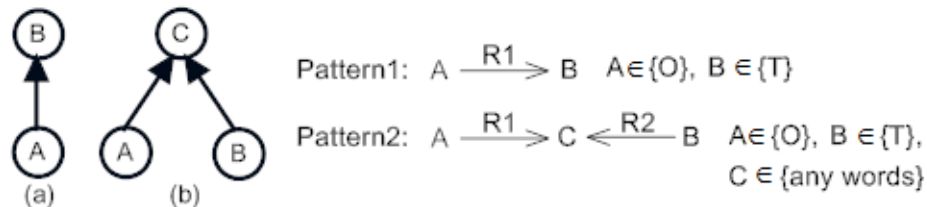
```

1: Calculate the seed alignment  $a_0$  using HMM model
2: while  $ic(a_0) > 0$  do
3:   if  $\{a : ic(a) < ic(a_0)\} = \emptyset$  then
4:     break
5:   end if
6:    $a_0 := \arg \max_{a \in nb(a_0), ic(a) < ic(a_0)} Pr(f|e, a)$ 
7: end while
8:  $M_{ij} := -1$  if  $(i, j) \notin \alpha_I^J$  or  $(i, 0) \in \alpha_I^J$ 
9: loop
10:   $S_{jj'} := -1$  if  $(j, a_{j'}) \notin \alpha_I^J$  or  $(j', a_j) \notin \alpha_I^J$ 
11:   $M_{i_1j_1} = \arg \max M_{ij} ; S_{j_1j'_1} = \arg \max S_{ij}$ 
12:  if  $M_{i_1j_1} \leq 1$  and  $S_{j_1j'_1} \leq 1$  then
13:    Break
14:  end if
15:  if  $M_{i_1j_1} > S_{j_1j'_1}$  then
16:    Update  $M_{i_1*}, M_{j_1*}, M_{*i_1}, M_{*j_1}$ 
      and  $S_{i_1*}, S_{j_1*}, S_{*i_1}, S_{*j_1}$ , set  $a_0 := M_{i_1j_1}(a_0)$ 
17:  else
18:    Update  $M_{j_1*}, M_{j'_1*}, M_{*j_1}, M_{*j'_1}$ 
      and  $S_{j'_1*}, S_{j_1*}, S_{*j'_1}, S_{*j_1}$ , set  $a_0 := S_{j_1j'_1}(a_0)$ 
19:  end if
20: end loop
21: Return  $a_0$ 

```

Obtain Partial Alignment Links

■ High-precise Syntactic Patterns



Pattern#1: $\langle OC \rangle \xrightarrow{mod} \langle TC \rangle$
Example: This phone has an <i>amazing design</i>
Pattern#2: $\langle TC \rangle \xrightarrow{obj} \langle OC \rangle$
Example: I <i>like</i> this <i>phone</i> very much
Pattern#3: $\langle OC \rangle \xrightarrow{pnmod} \langle TC \rangle$
Example: the <i>buttons easier</i> to use
Pattern#4: $\langle OC \rangle \xrightarrow{mod} (NN) \xleftarrow{subj} \langle TC \rangle$
Example: <i>IPhone</i> is a <i>revolutionary</i> smart phone
Pattern#5: $\langle OC \rangle \xrightarrow{pred} (VBE) \xleftarrow{subj} \langle TC \rangle$
Example: The <i>quality</i> of LCD is <i>good</i>

Label Propagation

- The confidence label can propagate among vertices

$$L_v^{i+1} = P_{con}(v) \times D_v^i + P_{inj}(v) \times I_v + P_{abnd}(v) \times L_\phi$$

$$D_v^i = \sum_u \frac{W(v, u)}{\sum_u W(v, u)} L_u^i$$

$P_{con}(v)$ The probability of continuing the random walk to the neighbors

$P_{abnd}(v)$ The probability of abandon the random walk

$P_{inj}(v)$ The probability of stop walk and emit a confidence according to the prior knowledge

$$P_{inj}(v) = \frac{j_v}{z_v}, \quad P_{con}(v) = \frac{c_v}{z_v}$$

$$P_{abnd}(v) = 1 - P_{con}(v) - P_{inj}(v)$$

$$c_v = \frac{\log(\beta)}{\log(\beta + \exp(H(v)))} \quad H(v) = - \sum_u p_{uv} \times \log(p_{uv})$$

$$j_v = (1 - c_v) \times \sqrt{H(v)} \quad p_{uv} = \frac{W(u, v)}{\sum_u W(u, v)}$$

$$z_v = \max(c_v + j_v, 1)$$

Experimental Results

Methods	Camera			Car			Laptop			Phone		
	P	R	F	P	R	F	P	R	F	P	R	F
Hu	0.63	0.65	0.64	0.62	0.58	0.60	0.51	0.67	0.58	0.69	0.60	0.64
DP	0.71	0.70	0.70	0.72	0.65	0.68	0.58	0.69	0.63	0.78	0.66	0.72
Zhang	0.71	0.78	0.74	0.69	0.68	0.68	0.57	0.80	0.67	0.80	0.71	0.75
Liu	0.75	0.81	0.78	0.71	0.71	0.71	0.61	0.85	0.71	0.83	0.74	0.78
Ours	0.77	0.82	0.79	0.74	0.71	0.72	0.66	0.85	0.74	0.85	0.75	0.80

Table 1: Experimental Results on *COAE 2008*

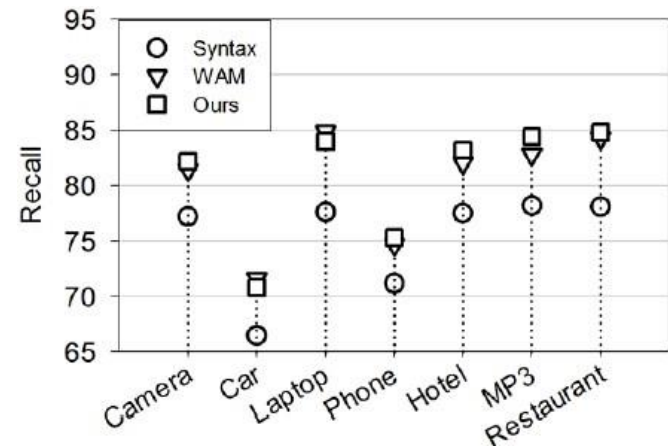
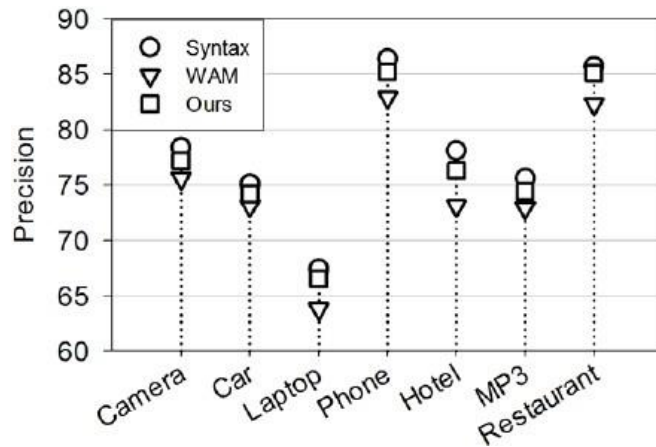
Methods	Hotel			MP3			Restaurant		
	P	R	F	P	R	F	P	R	F
Hu	0.60	0.65	0.62	0.61	0.68	0.64	0.64	0.69	0.66
DP	0.67	0.69	0.68	0.69	0.70	0.69	0.74	0.72	0.73
Zhang	0.67	0.76	0.71	0.67	0.77	0.72	0.75	0.79	0.77
Liu	0.71	0.80	0.75	0.70	0.82	0.76	0.80	0.84	0.82
Ours	0.76	0.83	0.79	0.74	0.84	0.79	0.85	0.85	0.85

Table 2: Experimental Results on *Large*

Methods	D1			D2			D3			D4			D5		
	P	R	F	P	R	F	P	R	F	P	R	F	P	R	F
Hu	0.75	0.82	0.78	0.71	0.79	0.75	0.72	0.76	0.74	0.69	0.82	0.75	0.74	0.80	0.77
DP	0.87	0.81	0.84	0.90	0.81	0.85	0.90	0.86	0.88	0.81	0.84	0.82	0.92	0.86	0.89
Zhang	0.83	0.84	0.83	0.86	0.85	0.85	0.86	0.88	0.87	0.80	0.85	0.82	0.86	0.86	0.86
Liu	0.84	0.85	0.84	0.87	0.85	0.86	0.88	0.89	0.88	0.81	0.85	0.83	0.89	0.87	0.88
Ours	0.86	0.84	0.85	0.88	0.83	0.85	0.89	0.90	0.89	0.81	0.83	0.82	0.91	0.87	0.89

Table 3: Experimental Results on *Customer Review Dataset*

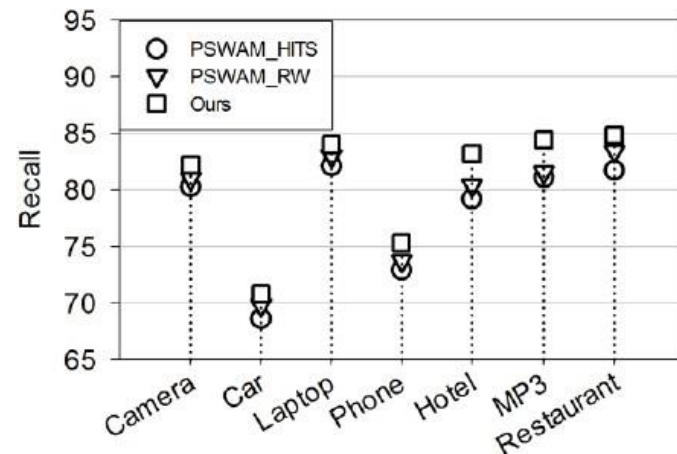
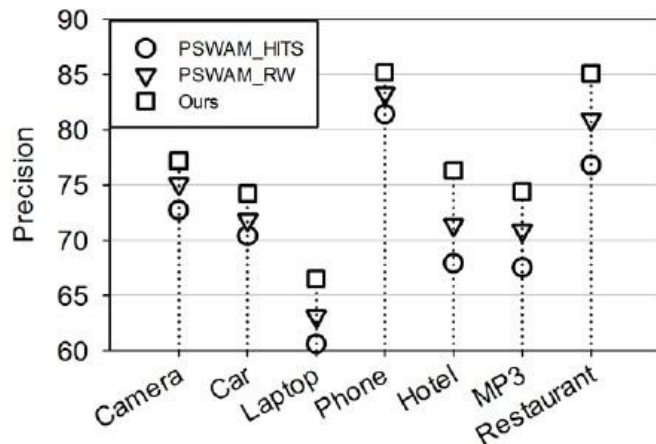
The Effectiveness of PSWAM



- Syntax : Syntactic patterns
- WAM : Standard word alignment model
- Ours : PSWAM

} + Label Propagation

The Effectiveness of our ranking algorithm



- PSWAM_HITS : PSWAM + HITS
- PSWAM_RW : PSWAM + Random Walk
- Ours : PSWAM + Label Propagation

Is WAM always useful in different sizes dataset

- Varying data size
 - 500→1,000,000

数据量大的时候，精准的opinion relation识别结果没有作用

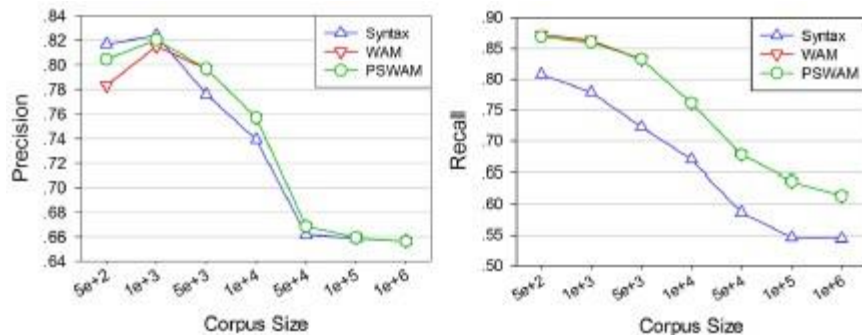


Figure 3: Experimental results on Restaurant

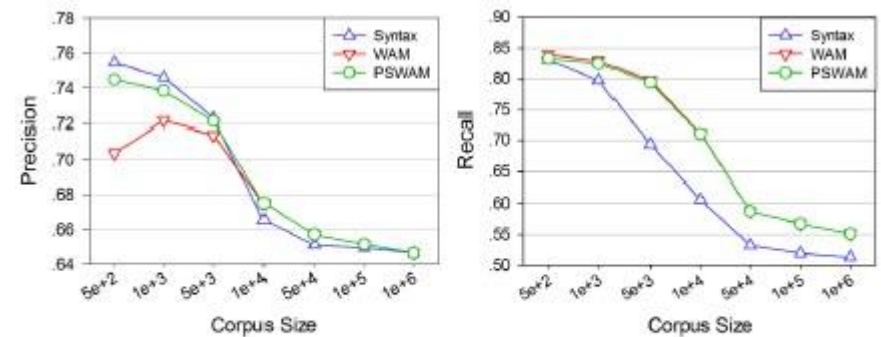


Figure 4: Experimental results on Hotel

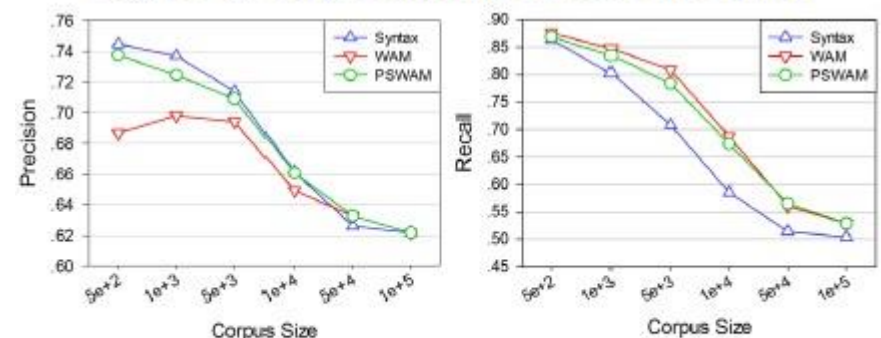
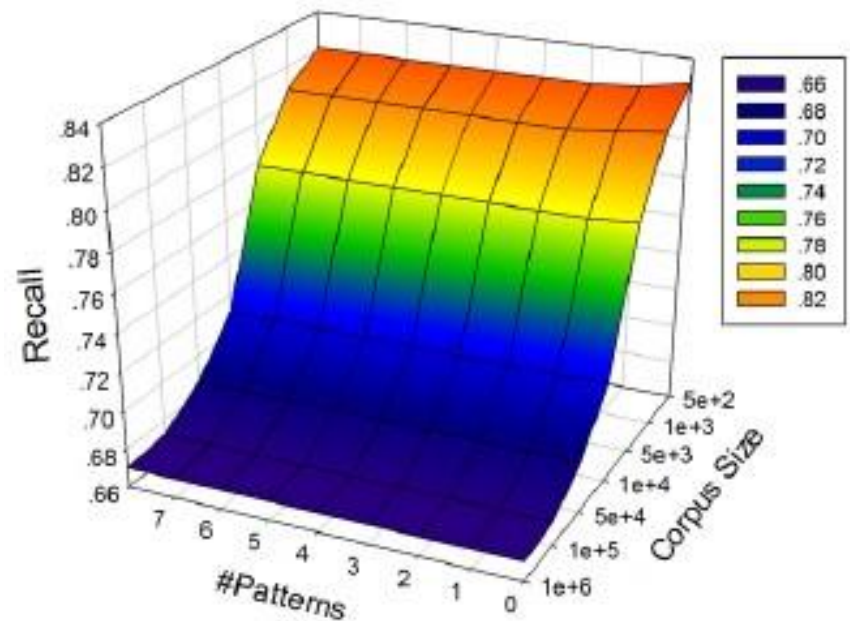
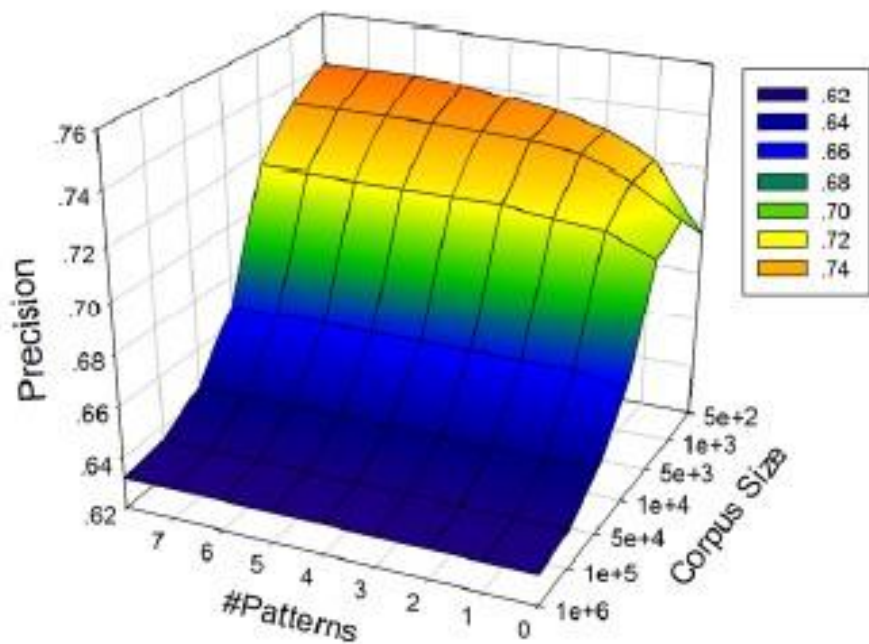


Figure 5: Experimental results on MP3

The Effect of Different Syntactic Information on WAM



Conclusions

- Graph-based approach is more useful than bootstrapping framework
 - Considering patterns in the graph is useful to improve precision
 - Word alignment model is effective for capturing opinion relations in sentences
 - Combing syntactic information with WAM is useful
 - Make penalty on the high-degree vertices is useful for getting rid of some incorrect target words
 - For big data, high-precise NLP techniques are useful?
-

论文发表

- Kang Liu, Liheng Xu and Jun Zhao, **Syntactic Patterns versus Word Alignment: Extracting Opinion Targets from Online Reviews**, in *Proceedings of ACL 2013*, Sofia, Bulgaria, August 4-9.
- Kang Liu, Liheng Xu and Jun Zhao, **Opinion Target Extraction Using Partial-Supervised Word Alignment Model**, in *Proceedings of IJCAI 2013*, Beijing, China, August 5-9.
- Kang Liu, Liheng Xu and Jun Zhao, **Opinion Target Extraction Using Word-Based Translation Model**, In *Proceedings of EMNLP-CoNLL 2012*, Jeju, Korea, July (oral).
- Liheng Xu, Kang Liu, Siwei Lai, Yubo Chen and Jun Zhao, **Mining Opinion Words and Opinion Targets in a Two-Stage Framework**, in *Proceedings of ACL 2013*, Sofia, Bulgaria, August 4-9.
- Liheng Xu, Kang Liu, Siwei Lai, Yubo Chen and Jun Zhao, **Walk and Learn: A Two-Stage Approach for Opinion Words and Opinion Targets Co-Extraction**, in *Proceedings of WWW 2013*, Rio de Janeiro, Brazil, May 13-17.

Q&A

Thanks